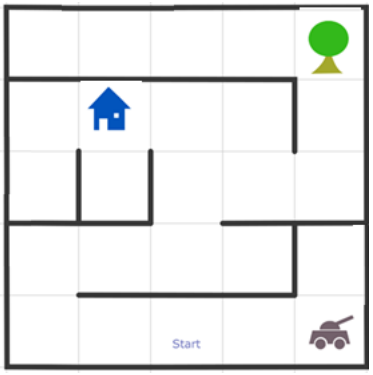
وزارة الدفاع
MINISTRY OF DEFENCE

DRONE CHALLENGE

Drone Challenge 2 – Object Detection in Maze

1. Competition Overview

Drone Challenge 2 – Object Detection in Maze is a technical drone competition focused on autonomous navigation, computer vision, and human–machine interface design. Teams must design and operate a custom GUI-based Command × Intelligence Interface (CxI) to control a drone and detect objects inside a maze.



2. Competition Inputs

On competition day, each team will be provided with:
- Maze size (NxN)
- Start location (x, y)
-Drone initial bearing (North, East, West, South)
- Object locations (x, y)
All provided inputs must be entered through the GUI. Hardcoding is strictly forbidden.

-A predefined AI model (The same model provided during the workshop)

3. Competition Phases

Discovery Phase (15 minutes):
- No scoring
- Teams explore the maze and gather information to be used in the Race Phase

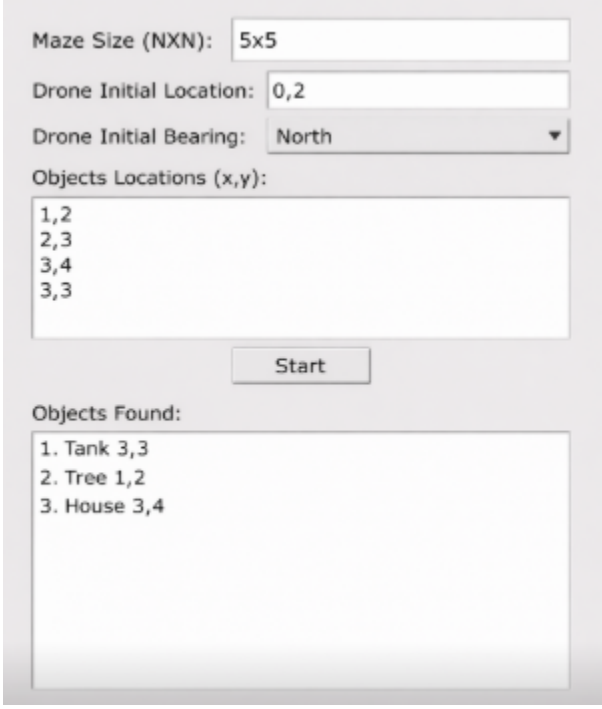
Race Phase (15 minutes):
- Teams detect and correctly identify all objects
- Time and accuracy determine ranking

4. CxI (GUI) Mandatory Requirements-For Race Phase

The GUI has “Start Mission” button to start the race phase.

The GUI must allow input of:
- Maze size
- Start location
- Object locations

The GUI must also display:
- Detected objects
- Snapshot of the detected object
- Object locations on the grid



Sample GUI

5. Object Detection & Validation

Each detected object triggers a snapshot that is shown to judges for validation. Objects must be correctly identified and localized to be counted.

6. Crash & Restart Policy

If the drone crashes or stops during the race phase:
- Restart occurs from the crash location
- Previously validated objects remain counted
- Time continues to run

7. Scoring System

Primary scoring is based on:
1. Number of correctly identified objects
2. Time to completion

Teams finding all objects are ranked by fastest time.

8. Code Integrity Rules

Code modification during the competition is strictly prohibited.
Only GUI input is allowed once the competition starts.
Violations result in immediate disqualification.

9. AI Model Rules

The provided AI model must be used as-is.
No retraining, fine-tuning, or replacement is allowed.

10. Judge Authority

Judges may inspect systems, verify GUI usage, and validate detections.